GRADE 10 CBE NOTES
COMPUTER SCIENCE

STEM PATHWAY

THIS PDF COMPRISES OF PART OF THE NOTES OF THE NAMED LEARNING AREA

FOR COMPLETE GRADE 10 NOTES:

CONTACT

LIST OF LEARNING AREAS AT SENIOR SCHOOL

Compulsory Subjects	Science, Technology, Engineering & Mathematics (STEM)	Social Sciences	Arts & Sports Science
1. English 2. Kiswahili/KSL 3. Community Service Learning 4. Physical Education *NB: ICT skills will be offered to all students to facilitate learning and enjoyment*	5. Mathematics/Advanced Mathematics 6. Biology 7. Chemistry 8. Physics 9. General Science 10. Agriculture 11. Computer Studies 12. Home Science 13. Drawing and Design 14. Aviation Technology 15. Building and Construction 16. Electrical Technology 17. Metal Technology 18. Power Mechanics 19. Wood Technology 20. **Media Technology*** 21. Marine and Fisheries Technology*	22. Advanced English 23. Literature in English 24. Indigenous Language 25. Kiswahili Kipevu/Kenya Sign Language 26. Fasihi ya Kiswahili 27. Sign Language 28. Arabic 29. French 30. German 31. Mandarin Chinese 32. History and Citizenship 33. Geography 34. Christian Religious Education/ Islamic Religious Education/Hindu Religious Education 35. Business Studies	36. Sports and Recreation 37. Physical Education (C) 38. Music and Dance 39. Theatre and Film 40. Fine Arts

COMPUTER SCIENCE GRADE 10 SENIOR SCHOOLS CONPLETE NOTES {STEM}

COMPETENCE BASED EDUCATION

SUMMARY OF STRANDS AND SUB STRANDS

1.0 FOUNDATION OF COMPUTER SCIENCE

1.1 Evolution of Computers

1.2 Computer Architecture

1.3 Input/Output (I/O) Devices

1.4 Computer storage

1.5 Central Processing Unit (CPU)

1.6 Operating System (OS)

1.7 Computer setup

2.0 COMPUTER NETWORKING

2.1 Data communication

2.2 Data Transmission Media

2.3 Computer Network Elements

2.4 Network Topologies

3.0: SOFTWARE DEVELOPMENT

3.1 Computer Programming Concepts

3.2 Program development

3.3 Identifiers and Operators

3.4 Control Structures

3.5 Containers

3.6 Functions

3.7

STRAND 1.0: FOUNDATION OF COMPUTER SCIENCE

SUB-STRAND 1.1: EVOLUTION AND DEVELOPMENT OF COMPUTERS

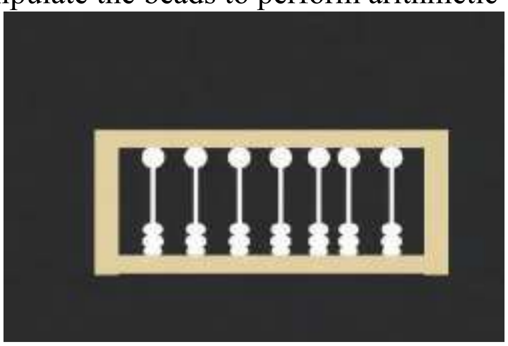
1.1.1 Early Computing Devices (Relating to Electronic Computers)

These early tools, though not electronic, introduced fundamental concepts in computation and paved the way for the development of modern computers.

(a) Ancient and Medieval Computing Devices:

- ✓ **Abacus (Around 2400 BC):**

Description: A manual calculating tool consisting of beads arranged on rods or wires, representing place values. Operators manipulate the beads to perform arithmetic operations.



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(Simplified Abacus Representation)
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An image showing a traditional abacus with beads on rods.

- **Relation to Evolution:** Introduced the concept of representing numbers using physical objects and performing calculations through manipulation, a basic form of data processing.

- ✓ **Napier's Bones (Early 17th Century):**
  - **Description:** A set of numbered rods used for multiplication and division. The rods contained precalculated multiples of numbers, simplifying complex calculations.

Image:

An illustration of a set of Napier's Bones.

- **Relation to Evolution:** Demonstrated the idea of using pre-calculated information and structured tools to simplify complex mathematical tasks, a step towards automation.

✓ **Pascaline (1642):**

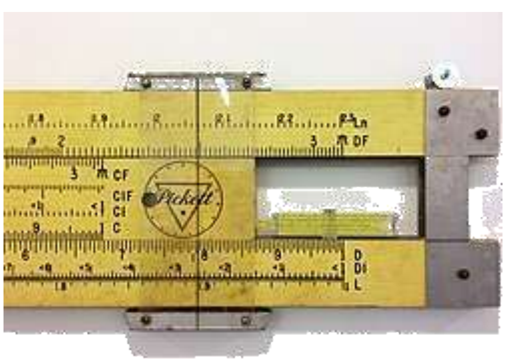
- **Description:** The first mechanical calculator capable of addition and subtraction. It used a system of gears and dials to represent numbers and perform calculations automatically.

A photograph of Blaise Pascal's mechanical calculator, the Pascaline.

- **Relation to Evolution:** Marked a significant step towards automated calculation using mechanical principles, a precursor to the electromechanical devices.

✓ **Slide Rule (Around 1620):**

- **Description:** An analog computer used primarily for multiplication and division, and also for functions such as roots, logarithms, and trigonometry. It uses two logarithmic scales that slide against each other.



Difference Engine (Early 19th Century):

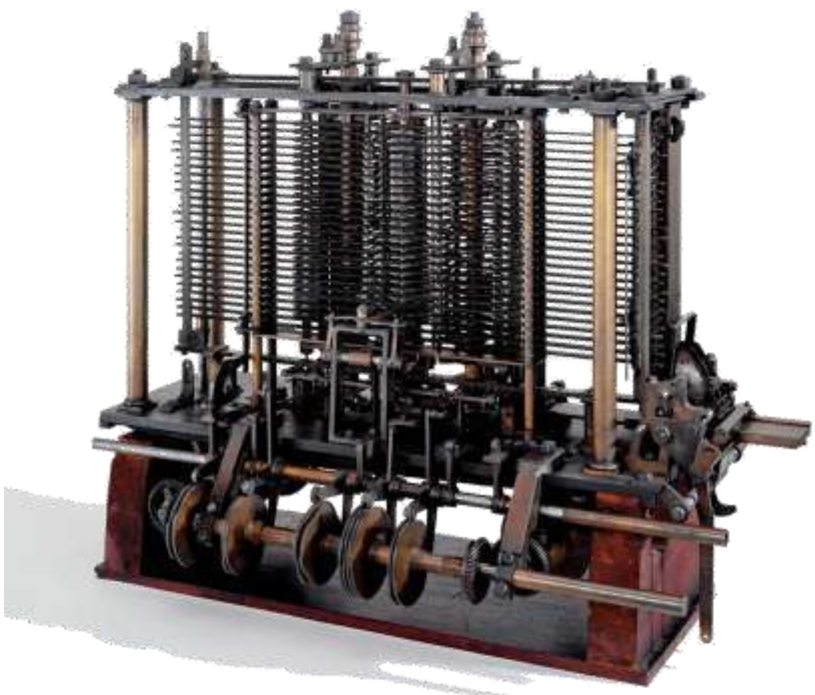
- **Description:** A mechanical calculator designed by Charles Babbage to automatically calculate polynomial functions. It was intended to produce accurate mathematical tables, eliminating errors from manual calculation.

Difference Engine No. 1.

- **Relation to Evolution:** Introduced concepts like automated computation based on a program (though fixed), mechanical memory for intermediate results, and the idea of a machine performing complex tasks autonomously.

✓ **Analytic Engine (Mid-19th Century):**

- **Description:** Also designed by Charles Babbage, this was a more ambitious general-purpose mechanical computer. It included components analogous to a modern computer: an arithmetic unit ("mill"), a control unit, memory ("store"), and input/output mechanisms (based on punched cards). Although never fully built in Babbage's lifetime, its design was revolutionary.

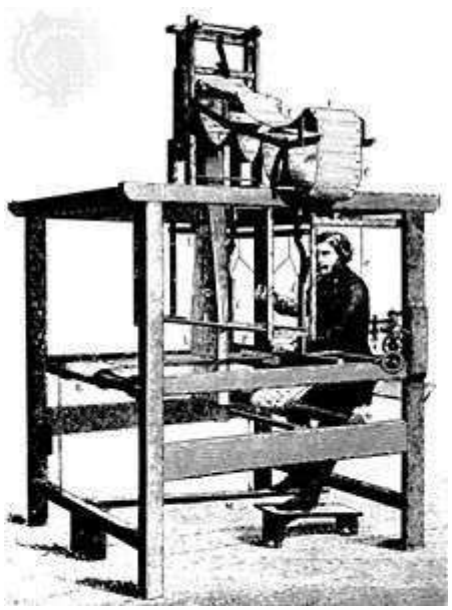


A diagram illustrating the conceptual architecture of the Analytic Engine.

- **Relation to Evolution:** Considered the theoretical precursor to the modern digital computer. It introduced the fundamental architectural concepts of input, processing, storage, and output, along with the idea of programmable computation. Ada Lovelace is considered the first computer programmer for her work on the Analytic Engine.

✓ **Jacquard Loom (Early 19th Century):**

- **Description:** A mechanical loom that used punched cards to control the weaving of intricate patterns in textiles automatically. The holes in the cards determined which warp threads were raised or lowered for each pass of the shuttle.



A close-up of the mechanism of a Jacquard Loom using punched cards.

- **Relation to Evolution:** Introduced the concept of using punched cards as a form of programmable instruction and data storage, which later influenced input methods for early computers.

1.1.2 Principle Technologies Defining Computer Development

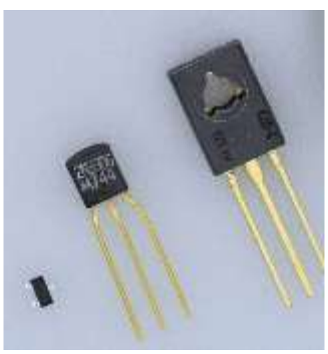
The evolution of computers is marked by breakthroughs in fundamental electronic technologies.

(b) Principle Technologies:

- **Vacuum Tubes (Early to Mid-20th Century):**
  - **Description:** Electronic devices that control the flow of electric current in a high vacuum between electrodes. They were used as switches and amplifiers in the first generation of electronic computers.

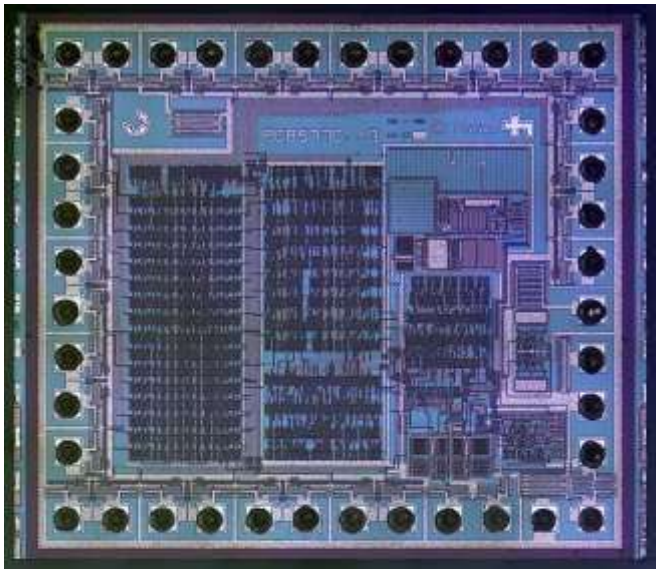
A schematic diagram of a triode vacuum tube.

  - **Impact:** Enabled the development of electronic computers by providing a way to perform calculations electronically. However, they were large, consumed a lot of power, generated significant heat, and were unreliable due to frequent burnouts.
- **Transistors (Mid-20th Century):**
  - **Description:** Semiconductor devices that can switch and amplify electronic signals. They replaced vacuum tubes due to their smaller size, lower power consumption, less heat generation, higher speed, and greater reliability.



A photograph of a small transistor.

  - **Impact:** Ushered in the second generation of computers, making them smaller, faster, more energyefficient, and more reliable. This was a crucial step towards the personal computer era.
- **Integrated Circuits (ICs) - Small Scale Integration (SSI) and Medium Scale Integration (MSI) (Late 20th Century):**
  - **Description:** A single semiconductor chip containing many interconnected electronic components (transistors, resistors, capacitors, etc.). SSI typically contained a few transistors, while MSI contained hundreds.



A photograph of a Dual In-line Package (DIP) integrated circuit.

  - **Impact:** Led to the third generation of computers. ICs further reduced the size, cost, and increased the speed and efficiency of computers. More complex circuits could be manufactured on a single chip, leading to more powerful machines.
- **Large Scale Integration (LSI) and Very Large Scale Integration (VLSI) (Late 20th Century - Present):**
  - **Description:** Advanced IC technologies that allowed for the integration of thousands (LSI) and then millions (VLSI) of transistors onto a single chip. Microprocessors (the entire CPU on a single chip) became possible with VLSI.

A photograph of an Intel 486DX2 microprocessor chip.

  - **Impact:** Revolutionized computing, leading to the development of powerful microprocessors, personal computers, laptops, and eventually the smartphones and tablets we use today. VLSI continues to drive miniaturization, increased processing power, and reduced costs in modern electronics (fourth and fifth generations).

1.1.3 Computer Generations and Principle Technologies

Each generation of computers is characterized by the dominant technology used in their construction.

(c) Relating Principle Technologies to Computer Generation:

Generation	Approximate Period	Principal Technology	Key Characteristics	Examples
First	1940s - 1950s	**Vacuum Tubes**	Large size, high power consumption, significant heat, low reliability, slow speed.	ENIAC, UNIVAC I
Second	1950s - 1960s	**Transistors**	Smaller size, lower power consumption, less heat, higher reliability, faster speed.	IBM 1401, PDP-1
Third	1960s - 1970s	**Integrated Circuits (SSI/MSI)**	Further reduction in size and cost, increased speed and efficiency, more reliable.	IBM System/360, DEC PDP-8
Fourth	1970s - Present	**Large Scale Integration (LSI) & Very Large Scale Integration (VLSI)**	Microprocessors, personal computers, high processing power, small size, low cost.	Apple Macintosh, IBM PC, Modern Laptops and Desktops
Fifth	Present and Beyond	**Artificial Intelligence, Parallel Processing, Superconductors, Quantum Computing**	Focus on AI, parallel processing, natural language understanding, advanced computing.	AI systems, Supercomputers, Quantum Computers (in development)

1.1.4 Technological Advancement in Computer Development

(d) Appreciating Technological Advancement:

The progression through computer generations demonstrates remarkable technological advancement:

- ✓ **Miniaturization:** Computers have shrunk from room-sized machines to handheld devices, thanks to the increasing density of integrated circuits.
- ✓ **Increased Processing Power:** Each new technology has enabled faster and more complex calculations, leading to the powerful computing capabilities we have today.
- ✓ **Improved Efficiency:** Newer technologies consume less power and generate less heat, making computers more energy-efficient and reliable.

- ✓ **Reduced Cost:** Mass production of advanced components like microprocessors has significantly lowered the cost of computing, making it accessible to a wider population.
- ✓ **Enhanced Reliability:** Solid-state components like transistors and integrated circuits are far more reliable than their predecessors, leading to fewer hardware failures.
- ✓ **New Capabilities:** The advancements have paved the way for entirely new applications and fields, such as the internet, multimedia, artificial intelligence, and data science.

SUB-STRAND 1.2: COMPUTER ORGANISATION AND ARCHITECTURE

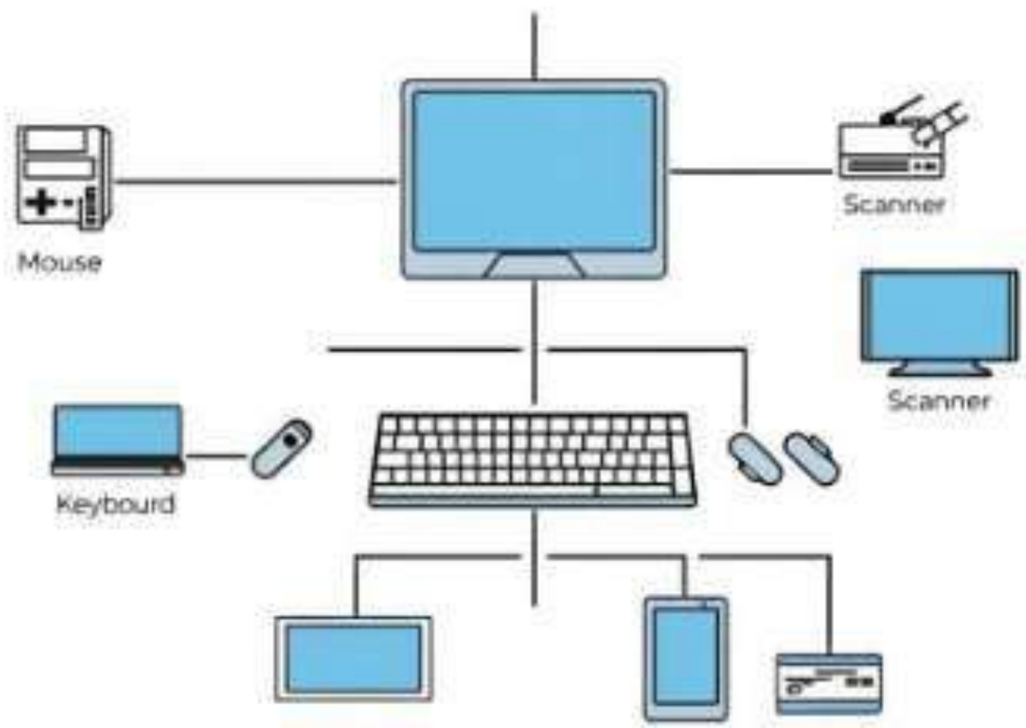
1.2.1 Von Neumann Computer Architecture

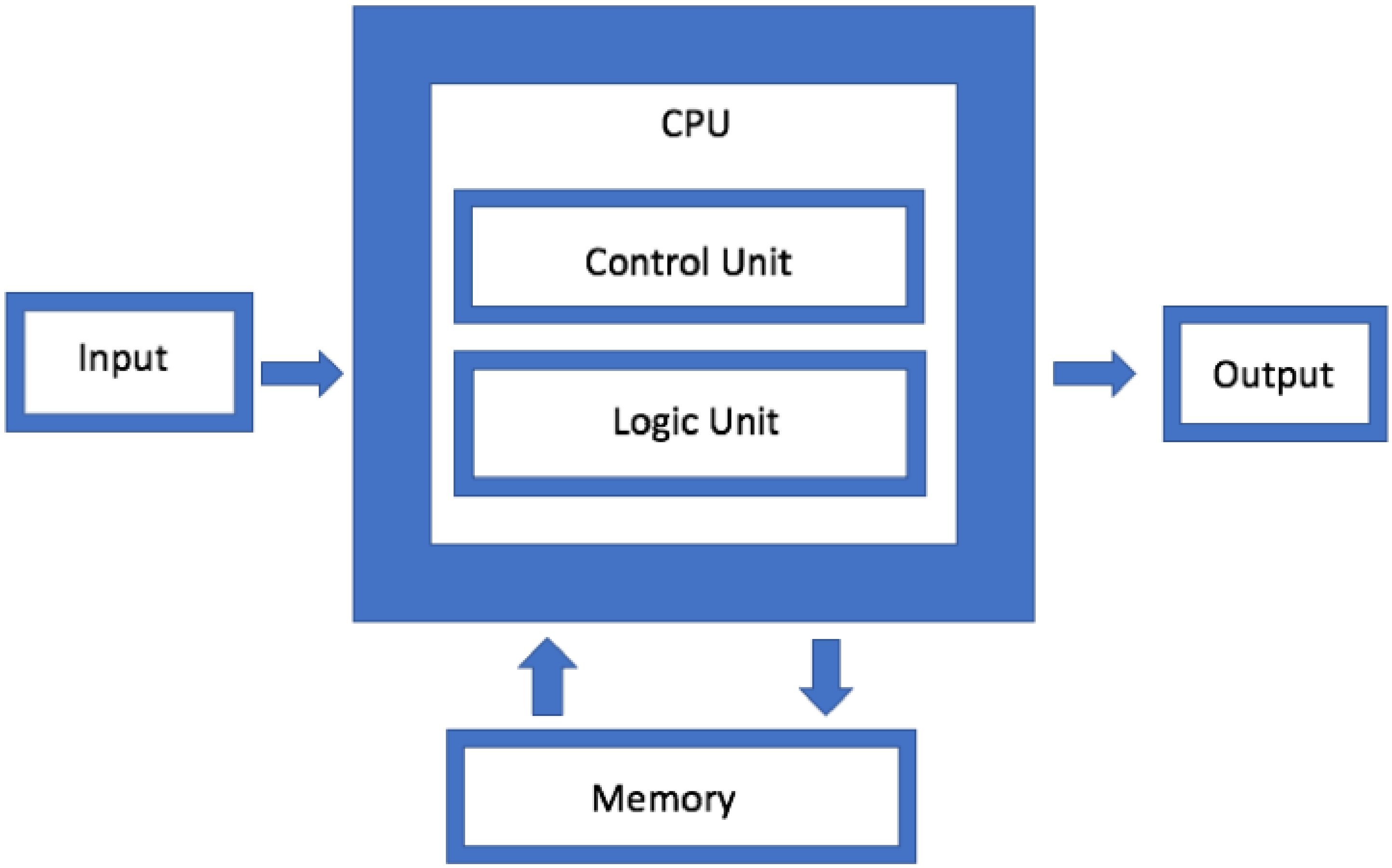
The von Neumann architecture is a computer architecture based on a description by the mathematician and physicist John von Neumann and others in the "First Draft of a Report on the EDVAC" (1945). It is the foundation of most modern computers.

(a) Functional Organisation of a von Neumann Computer:

A von Neumann architecture has five main functional units:

1. **Input Unit:** Devices used to feed data and instructions into the computer. Examples include the keyboard, mouse, scanner, microphone, etc.





A diagram showing various input devices.

2. **Central Processing Unit (CPU):** The "brain" of the computer, responsible for executing instructions. It consists of two main parts:
   - ✓ **Arithmetic Logic Unit (ALU):** Performs arithmetic calculations (addition, subtraction, multiplication, division) and logical operations (AND, OR, NOT).
   - ✓ **Control Unit (CU):** Manages and coordinates all the activities of the computer. It fetches instructions from memory, decodes them, and sends control signals to other units to execute them.

LIST OF GRADE 10 SENIOR SCHOOLS MATERIALS AVAILABLE:

1	**TEACHING/LEARNING NOTES**
2	**KICD APPROVED TEXTBOOKS**
3	**KICD APPROVED TEACHER GUIDES**
4	**REVISION BOOKS**
5	**SCHEMES OF WORK**
6	**LESSON PLANS**
7	**KICD APPROVED CURRICULUM DESIGNS**
8	**RECORD OF WORK**
9	**EXAMINATIONS & ASSESSMENT MATERIALS**
10	**APPROVED PRACTICAL BOOKS & PRACTICAL MATERIALS**
12.	**APPROVED LITERATURE & FASIHI BOOKS & ALL RELATED MATERIALS**